

Active Rehabilitation and Return to Play in Sports-Related Concussion



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KEYWORDS

- Return to play • Return to learn • Active rehabilitation
- Subsymptom threshold exercise

KEY POINTS

- Active rehabilitation is safer and more effective than prolonged rest: early, progressive exercise accelerates recovery can reduce symptom burden, with subthreshold intensity producing the largest benefit.
- Individualized, threshold-guided exercise prescriptions: use objective thresholds to determine safe exercise intensity and progression, starting with light activity and advancing as symptoms permit.
- Graduated return-to-learn (RTL) and return-to-play (RTP): RTL should precede RTP in youth; symptom absence is not necessary to begin RTP, careful supervision is essential.
- Recovery trajectories are variable and influenced by multiple factors: most adults recover in ~2 weeks, youth up to ~4 weeks, but injury and pre-injury factors can prolong recovery beyond 28 days.
- Targeted and multidisciplinary management for persistent symptoms: multimodal rehab, emphasis on psychological readiness and school accommodations, and careful consideration of medications.

INTRODUCTION

Sport-related concussion (SRC) represents a complex pathophysiological process and significant public health challenge, affecting approximately 1.6 to 3.8 million athletes in the United States annual.¹ The American Congress of Rehabilitation Medicine defines concussion as a mild traumatic brain injury (TBI), characterized by an absence of macroscopic neural damage on neuroimaging and does not rely on significance of symptoms for classification.² The management of SRC has shifted over the past 2 decades, moving away from the historical approach of strict cognitive and physical rest, or *cocooning*, toward a more active, individualized rehabilitation strategy.³ Old

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Abbreviations	
CISG	Concussion in Sport Group
RTL	return to learn
RTP	return to play
TBI	traumatic brain injury

recommendations emerged from concerns about secondary impact syndrome and the role of the neurometabolic cascade following brain injury.^{4,5} Athletes with concussion were prescribed complete rest until all symptoms resolved, a practice that was largely empiric and lacked robust scientific support. The rationale for strict rest was based primarily on animal studies suggesting that forced early exercise after TBI could worsen neurometabolic dysfunction.⁶ However, these findings did not account for voluntary, controlled activity in humans, nor did they consider the physiologic and psychological consequences of prolonged inactivity. Enforced strict rest has since been associated with slower recovery and increased risk of persistent symptoms.^{7,8} Contemporary guidelines now emphasize early, symptom-limited activity and individualized, progressive rehabilitation protocols that balance therapeutic exercise with adequate recovery.^{9–11}

Understanding the underlying pathophysiology has informed the development of active rehabilitation strategies. Current evidence supports early mobilization, sub-threshold aerobic exercise, and targeted interventions based on specific symptom profiles.^{3,12,13} This approach is based on the understanding that concussion involves a complex neurometabolic cascade affecting cerebral blood flow, autonomic function, and cellular energy metabolism, processes that may benefit from controlled physical activity rather than complete rest.⁵ This article reviews the current evidence for active rehabilitation in SRC, examining recovery trajectories, targeted interventions, and graduated return-to-play (RTP) protocols. The evolution from strict rest to early, individualized, symptom-guided activity represents a paradigm shift in concussion management, supported by high-quality systematic reviews, randomized controlled trials, and consensus statements from leading sports medicine organizations.^{3,9–11,14,15}

RECOVERY TRAJECTORIES AND PREDICTORS

Recovery timelines from SRC can vary depending on multiple individual and preinjury/injury-related factors.^{16–18} While most adults recover within approximately 2 weeks, children and adolescents may require up to 4 weeks for full symptom resolution and return to unrestricted activity.¹⁹ In a recent systematic review and meta-analysis of recovery trajectories, the mean time to symptom resolution was 14.0 days (95% CI: 12.7, 15.4; n = 35 studies, I² = 98%, Q statistic < .01), with most returning to school by 8.3 days (95% CI: 5.6–11.1; n = 11 studies, I² = 99.3%, Q statistic < .01) and 93% back in school without new academic supports by 10 days.²⁰ A prospective observational study of 1751 collegiate athletes, participants began the RTP protocol within a median of 6.4 days after concussion and spent a median of 12.8 days in the protocol, with 85% cleared for unrestricted sport participation by 1 month postinjury.²¹

The initial symptom burden at presentation is one of the strongest and most consistent predictor of recovery duration. Patients presenting with higher total symptom scores or more severe individual symptoms are significantly more likely to experience prolonged recovery beyond 30 days.²² Recent data from collegiate athletes demonstrate that those endorsing symptoms across multiple domains, particularly when all

symptom domains are high, have substantially longer recovery times compared to those with isolated symptom clusters.²³

Several other modifying factors can significantly influence recovery trajectories. Adolescent age, female sex, history of migraine/headache, pre-existing depression or anxiety, and delayed access to medical care are associated with prolonged symptom duration.²⁴ Female athletes consistently demonstrate higher symptom burden and longer recovery times compared to male counterparts in comparable sports, a finding observed across multiple epidemiologic studies.^{25,26} This sex difference may reflect biological factors, reporting behaviors, or both, and warrants consideration in individualizing rehabilitation protocols. Athletes with attention-deficit/hyperactivity disorder or learning disabilities also report higher concussion incidence rates and may experience more complicated recovery courses.²⁷

History of previous concussion represents another important prognostic factor, though the relationship is complex.^{21,28} While multiple previous concussions may prolong recovery in some athletes, particularly if the interinjury interval is short, the presence of a prior concussion does not uniformly predict poor outcomes. Interestingly, neurometabolic studies suggest that prior concussion history may attenuate certain physiologic responses, such as myoinositol elevation, during subsequent injuries.²⁹ This finding highlights the need for individualized assessment rather than blanket restrictions based solely on concussion history.

Understanding these recovery trajectories and predictive factors is essential for clinicians to provide accurate prognostic information, identify athletes requiring closer monitoring or early specialist referral, and tailor rehabilitation programs to individual patients. Recognition that most athletes recover within a predictable timeframe also helps combat unnecessary anxiety while maintaining appropriate vigilance for those at higher risk of complications. However approximately 10% to 20% of individuals with SRC experience persistent symptoms beyond 28 days, necessitating more intensive, multidisciplinary management.¹⁹

EARLY ACTIVE REHABILITATION: SAFETY AND EFFICACY

Early, progressive aerobic activity initiated within 24 to 72 hours postinjury is safe and accelerates recovery compared to prolonged rest or delayed activity initiation. The evidence supporting active rehabilitation over prolonged rest has grown in recent years, reshaping concussion management guidelines. A systematic review and meta-analysis demonstrated that physical activity interventions significantly reduce symptom burden after concussion, with subthreshold aerobic exercise producing the largest effect size. When pooled across all studies, physical activity yielded an overall large effect on symptom reduction ($g = 1.03$), providing compelling evidence for its therapeutic benefit.³⁰ In a randomized controlled trial involving adolescents with SRC, those prescribed subthreshold aerobic exercise beginning 2 to 10 days postinjury achieved faster median recovery (13 days vs 17 days, $P = .009$) and had significantly fewer persistent symptoms at 28 days (21% vs 32%, $P = .04$) compared to those assigned to placebo-like stretching exercises. Importantly, this early exercise prescription did not cause harm or prolong symptoms in any subgroup, including those with initially severe symptoms.³¹

The rationale for early aerobic exercise stems from its effects on the pathophysiological disturbances following concussion. Exercise appears to restore autonomic nervous system function, improve cerebral blood flow regulation, and address the metabolic crisis characteristic of the acute postconcussion period.^{32,33} Aerobic exercise treatment alleviates emotional and vestibular symptoms as much as it does

headache or fatigue. While cognitive and noncontact physical exertion might increase symptoms temporarily, it is unlikely to worsen the underlying injury or long-term outcomes.

The exercise prescription is modeled after cardiac rehabilitation principles.³⁴ The heart rate at which patients experience an increase of more than 2 points in the severity of concussion symptoms at any stage during testing or treatment can be used as a threshold at which physical activity progression should be paused. Light activity such as activities of daily living and walking begun within 24 to 72 hours after injury facilitates recovery more effectively than resting until symptoms resolve. The Buffalo Concussion Treadmill Test has been developed as both a diagnostic tool and a means of prescribing individualized exercise intensity, using symptom exacerbation as the threshold for determining appropriate activity levels, and demonstrates excellent sensitivity (94.4%; 95% CI: 90.8–97.2) and specificity (94.6%; 95% CI: 91.1–97.3) for diagnosing SRC when performed within 14 days of injury.³⁵

SYMPTOM-SPECIFIC AND TARGETED INTERVENTIONS

While aerobic exercise forms the foundation of active rehabilitation, targeted interventions are beneficial for specific symptom profiles and persistent symptoms. For adolescents with persisting symptoms beyond 30 days, active rehabilitation and collaborative care may decrease symptom burden. A systematic review of 12 randomized controlled trials found that individually tailored, multimodal interventions combining cervical, vestibular, and oculomotor therapies were particularly effective for those with persistent symptoms, with concussed individuals approximately 3 times more likely to have returned to sport by 8 weeks if they received specific multimodal interventions.³⁶

Cervicovestibular rehabilitation is recommended for adolescents and adults with dizziness, neck pain, and/or headaches persisting beyond 10 days. A systematic review found that individualized cervicovestibular rehabilitation decreased time to return to sport compared with rest followed by gradual exertion (HR 3.91, 95% CI: 1.34–11.34) and when compared with a subtherapeutic intervention (HR 2.91, 95% CI: 1.01–8.43).¹⁰ Patients may respond favorably when a treatment plan is specifically designed to address deficits identified by a detailed vestibular/balance evaluation, including poor gaze stabilization, increased motion sensitivity, poor functional balance and gait measures, and cervicogenic dysfunction.

RETURN TO PLAY: PRINCIPLES AND FRAMEWORK

The graduated RTP protocol outlines the gradual transition to practice and competition after SRC.¹⁴ RTP traditionally was recommended when athletes reached their baseline level of symptoms, cognition, and balance/postural stability, but absolute symptom resolution may not be necessary before initiating a graduated return-to-sport protocol. The decision to RTP should be made by a qualified health care provider with expertise in concussion management. RTP follows a stepwise progression after an athlete has reached preinjury level of symptoms and function. Return to learn (RTL) should precede final clearance for RTP in pediatric and adolescent populations.³⁷

GRADUATED RETURN-TO-SPORT PROGRESSION

The graded RTP protocol has evolved with each iteration of the Concussion in Sport Group (CISG) recommendations. It is widely accepted but empiric, without evidence to support either the progression sequence or the time spent in each stage.

The stepwise progression outlined by CISG recommends the following stages:

1. Symptom-limited activity: Daily activities that do not provoke symptoms
2. Aerobic exercise:
 - 2A. Light: up to approx. 55% max HR
 - 2B. Moderate: up to approx. 70% max HR

Walking or stationary cycling at slow-to-medium pace; may start light resistance training. Goal is to increase heart rate.
3. Individual sport-specific exercise: Running or training drills; no activities with risk of head impact. Goal is to add movement and change of direction.
4. Noncontact training drills: Harder drills (eg, passing drills and team drills). Goal is exercise, coordination, and increased thinking during sport.
5. Full-contact practice: After medical clearance, participate in full, normal training activities. Goal is to restore confidence and allow coaching staff to assess functional skills.
6. Return to sport: Normal game play with full clearance/participation.³⁸

TIMING AND SYMPTOM MANAGEMENT DURING PROGRESSION

The return-to-sport progression should be individualized and take into account the injury, the athlete's age, history of SRC and level of play, and the ability to provide close supervision during the return to activity. RTP progression should ideally be monitored by a licensed health care professional who is knowledgeable in concussion management. Each stage should last at least 24 hours without worsening of symptoms before progressing to the next stage. In general, for young athletes, this 24-h minimum is particularly important to ensure adequate recovery between stages. The typical time frame for completing the entire progression is at least 1 week, though most patients diagnosed with concussion can expect to be withheld from competition for at least this duration. If symptoms return while the athlete is in the program, the athlete should wait 24 hours, and if the symptoms resolve, attempt the previous step that was completed without symptoms and continue the progression if symptoms do not recur. A return or worsening of symptoms may indicate incomplete recovery from the concussion. Re-evaluation by a health care provider is indicated for any athlete who has a continued return of symptoms with exertion.³⁸

Athletes should not advance beyond the initial stages until they return to their pre-injury symptom levels. However, recent evidence suggests that a range of symptom severity reporting exists at the time of graduated RTP protocol initiation. In a large cohort study of 1531 collegiate athletes, 90% had symptom severity scores of 0 to 5 when initiating the protocol, while 10% had scores above 5. Symptom severity scores at protocol initiation did not significantly differ in total time to unrestricted RTP for scores up to 10, though scores above 10 were associated with longer protocol duration.³⁹ These findings suggest that initiation of RTP before an absolute absence of all symptoms is not associated with longer progression, though clinicians should balance active recovery with avoidance of premature return to activity.

PROHIBITION OF SAME-DAY RETURN TO PLAY

The phrase "When in doubt, sit them out!" is paramount in the management of pediatric and adolescent concussion.⁴⁰ Despite the existence of state laws and multiple professional societies having published statements recommending not returning to play on the same day of the injury, literature reveals that 10% to 38% of young athletes reported returning to play on the same day as the injury.⁴¹ Athletes who continued to play immediately after SRC were found to have worse symptoms and cognitive scores

than those who were removed immediately from play. These athletes were also 8.8 times more likely to have a recovery longer than 21 days.⁴² Another study demonstrated that athletes who sustained an additional head impact within 24 hours of the first had a greater symptom burden and a longer recovery time (52.3 vs 36.9 days) than those who did not sustain an additional head injury.⁴³ These studies reinforce the importance of immediate removal from play to reduce the likelihood of a longer recovery and worse symptoms, or exposure to additional head trauma.

RETURN TO LEARN AND SCHOOL INTEGRATION SHOULD PRECEDE RETURN TO PLAY IN YOUTH

RTL represents a critical component of SRC management, particularly for pediatric and adolescent athletes. Concussion management guidelines now prioritize return to school over return to sport, recognizing that cognitive symptoms can significantly impede academic performance and that successful school reintegration must precede full athletic participation.³⁷ Recent observational data demonstrate that early cognitive activity, including an early return to school, is associated with faster recovery, whereas prolonged absences from school and other life activities may delay recovery.⁴⁴

RTL protocols should be individualized and symptom-guided, following a stepwise progression similar to RTP protocols. Medical and school-based teams should counsel students and families regarding the process of gradually increasing the duration and intensity of academic activities as tolerated, with the goal of increasing participation without significantly exacerbating symptoms. The most recent CISC Consensus Statement recommends a brief 24 to 48 hour rest period, during which students can engage in light activities, such as 5 to 15 minutes of screen time or reading, that do not exacerbate symptoms beyond mild discomfort. The typical progression includes: (1) daily activities that do not result in more than mild symptom exacerbation; (2) school activities such as homework and reading outside the classroom; (3) return to school part-time with gradual introduction of schoolwork and access to rest breaks; and (4) return to school full-time with gradual progression until a full day can be tolerated.⁴⁵

Academic accommodations are frequently necessary during recovery. Studies indicate that 17% to 73% of students were provided academic accommodations or experienced difficulty with return to school after concussion.⁴⁶ For students with prolonged symptoms that interfere with academic performance, school-based teams should assess educational needs and determine the need for additional educational supports. Educational supports should be monitored and adjusted on an ongoing basis by the school-based team until the student's academic performance has returned to preinjury levels.

PSYCHOLOGICAL READINESS

Beyond physical symptom resolution, psychological readiness represents a critical component of safe return to sport that must be assessed before final clearance. The athlete should demonstrate psychological readiness for RTP, recognizing that fear of reinjury, anxiety about performance, and concerns about long-term health consequences may affect an athlete's confidence and willingness to return to full-contact sport.⁴⁷ Athletes who harbor significant anxiety about returning to sport may benefit from additional counseling or gradual exposure to sport-specific scenarios before full clearance. Premature return in the absence of psychological readiness may increase injury risk or negatively impact performance.⁴⁸

PHARMACOLOGIC THERAPY CONSIDERATIONS

Pharmacologic therapy prescribed for management of SRC symptoms must be considered carefully during RTP. The decision to progress while on newly prescribed medication can depend on the benefits of symptom management versus potential effects on performance, safety, and the athlete's ability to accurately report symptoms. Medications commonly prescribed for concussion-related symptoms include analgesics for headache, medications for sleep disturbance, and treatments for mood disorders. Medications may mask symptoms that would otherwise indicate incomplete recovery. Additionally, medication can affect cognitive or physical performance in ways that could increase injury risk, potentially affecting safe participation in contact activities while taking the medication.

For athletes requiring ongoing pharmacologic therapy, the physician should ensure that symptoms are adequately controlled, that the athlete demonstrates stability on the medication regimen, and that the medication does not impair the athlete's ability to protect themselves during sport participation. Regular reassessment is essential to determine whether symptoms persist due to incomplete recovery or represent a chronic condition requiring ongoing management.

MULTIPLE CONCUSSIONS OR HISTORY OF PROLONGED RECOVERY

Athletes with a history of multiple concussions or prolonged recovery (over 4 weeks) require consideration of a more conservative approach to RTP decision-making. An athlete who has a history of multiple concussions or who had a prolonged recovery may need a longer period of time to progress through each step of the graduated RTP. Athletes with multiple prior concussions may demonstrate cumulative effects, including longer recovery times, increased symptom burden, and potentially increased vulnerability to subsequent injury.⁴⁹ For these complex cases, a specialist with experience in concussion management may be needed to create an appropriate return-to-activity program. The specialist can provide individualized assessment, identify specific deficits requiring targeted intervention, and develop a customized progression that accounts for the athlete's unique history and risk factors. An extended RTP progression may be necessary if the patient is held out for an extended amount of time and requires reconditioning for sport participation.⁵⁰

MONITORING AND FOLLOW-UP

Serial clinical assessment and symptom tracking are essential components of concussion management. Postconcussion symptoms and academic progress in school should be monitored collaboratively by the student, family, health care professionals, and school teams, who jointly determine what modifications or accommodations are needed to maintain an academic workload without significantly exacerbating symptoms. Criteria for specialist referral include persistent symptoms beyond expected recovery timeframes (typically 4 weeks for children and adolescents, 2 weeks for adults), complex cases with multiple comorbidities, and students who demonstrate prolonged symptoms and academic difficulties despite an active treatment approach. Communication with all stakeholders—including the athlete, family, coaches, athletic trainers, school personnel, and other health care providers—ensures coordinated care and appropriate support throughout recovery.

FUTURE DIRECTIONS AND RESEARCH GAPS

Despite significant advances in SRC management, substantial research gaps remain. Optimization of exercise timing, intensity, duration, and modality represents a critical research priority. While current evidence supports early, progressive aerobic exercise, further research is warranted to delineate how to optimize these parameters for different endophenotypes of SRC.

CLINICS CARE POINTS

- Pearl: The initial symptom burden is the strongest predictor of prolonged recovery; high early symptom loads should prompt closer follow-up and earlier multidisciplinary referral.
- Pearl: Prescribe early, controlled aerobic exercise within 24-48 hours of concussion rather than strict rest. Prolonged "cocooning" does not improve outcomes and may delay recovery.
- Pitfall: Do not require complete symptom resolution before initiating physical activity. Post-concussive symptoms may overlap with pre-existing symptoms (eg, fatigue, sleep problems), and activity can safely advance as long as exacerbation is ≤ 2 points on a 0-10 scale and resolves within an hour.
- Pitfall: Failing to assess the cervical spine and vestibular system is a common oversight. Cervicogenic headache and dizziness mimic persistent post-concussive symptoms, and cervicovestibular rehabiliy dramatically improves clearance rates.
- Pitfall: Allowing return to contact sport before full return to school, particularly in adolescents, violates current consensus guidelines; academic reintegration should always precede unrestricted sport participation.

DISCLOSURE

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